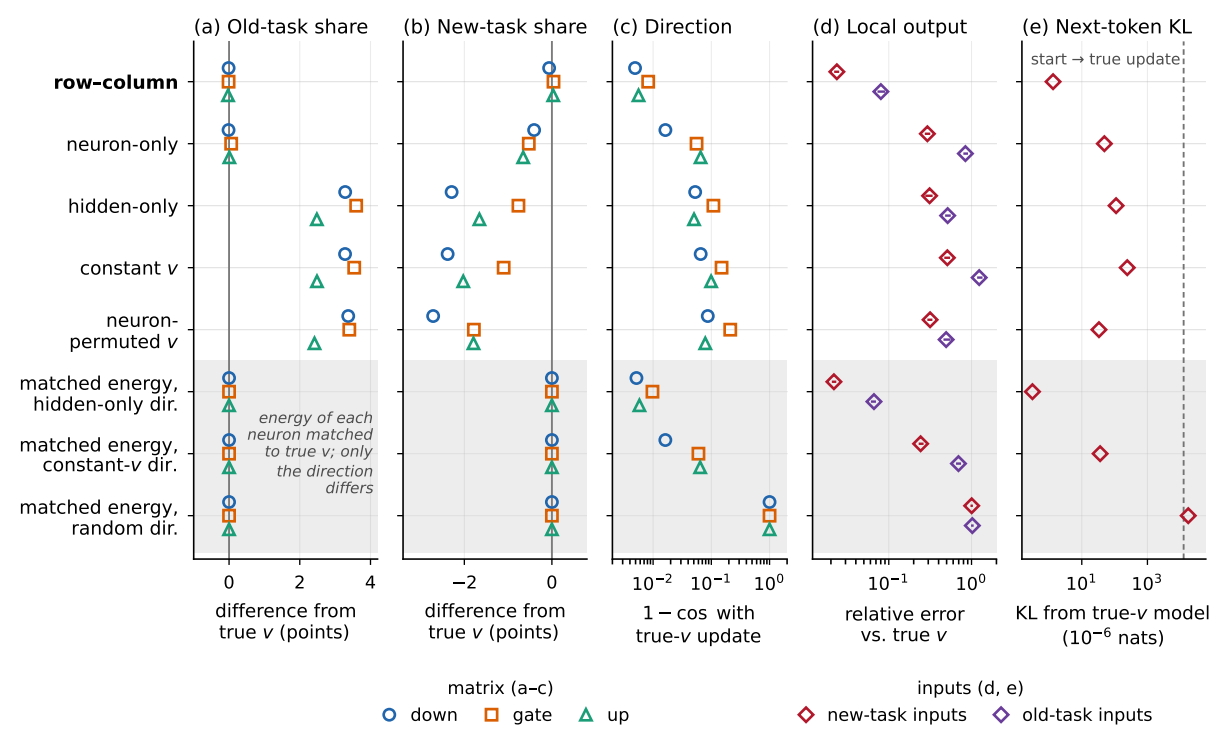
(a) Old-task share
(b) New-task share
(c) Direction
(d) Local output
(e) Next-token KL
row–column
neuron-only
hidden-only
constant v
neuron-permuted v
matched energy, hidden-only dir.
matched energy, constant-v dir.
matched energy, random dir.
energy of each neuron matched to true v; only the direction differs
start → true update
0
2
4
−2
0
10^{-2}
10^{-1}
10^{0}
10^{-1}
10^{0}
10^{1}
10^{3}
difference from true v (points)
difference from true v (points)
1 − cos with true-v update
relative error vs. true v
KL from true-v model (10^{-6} nats)
matrix (a–c)
down
gate
up
inputs (d, e)
new-task inputs
old-task inputs